

Project Proposal

Autonomous Delivery Cart

Date: 19/9/2025

1. Introduction

The Autonomous Delivery Cart project aims to design and implement a self-driving delivery solution for compounds, hotels, and gated communities. The system integrates **cloud computing, mobile applications, IoT communication, and AI-powered navigation** to provide a contactless, reliable, and efficient delivery service.

This project will be developed as a **graduation project** over two semesters, with the potential to evolve into a **startup** after completion.

2. Problem Statement

Manual delivery systems are slow, labor-intensive, and prone to errors. Current solutions lack real-time tracking, automation, and customer interactivity. Our project addresses these challenges by introducing an **autonomous, cloud-connected cart** that enhances delivery efficiency, reliability, and customer experience.

3. Project Objectives

- Develop a fully functional **cloud-based backend** using AWS for real-time communication.
- Build a **mobile application** for customers to place and track orders.
- Provide a **web app for vendors** (supermarkets, pharmacies) to manage orders and dispatch deliveries.
- Design a **manager dashboard** with analytics and monitoring tools.
- Integrate **IoT-enabled cart hardware** (GPS, sensors, camera, LiDAR, Kinect, ZED).
- Ensure scalability, security, and cost efficiency.

4. Scope of Work

Phase 1 (Semester 1 – 14 Weeks)

- Requirement gathering, analysis, and documentation.
- High-level system architecture design.
- Cloud infrastructure setup on AWS.
- Implementation of APIs, database schema, and IoT integration.
- MVP version of the mobile app and vendor web app.
- Documentation and diagrams for submission.

Phase 2 (Semester 2 – 14 Weeks)

- Advanced dashboard for managers (analytics, reporting).
- Full vendor marketplace web app.
- Data analytics pipeline for insights.
- AI-powered features (recommendations, navigation optimization).
- System scalability improvements (multi-cart support).
- Final deployment and testing.

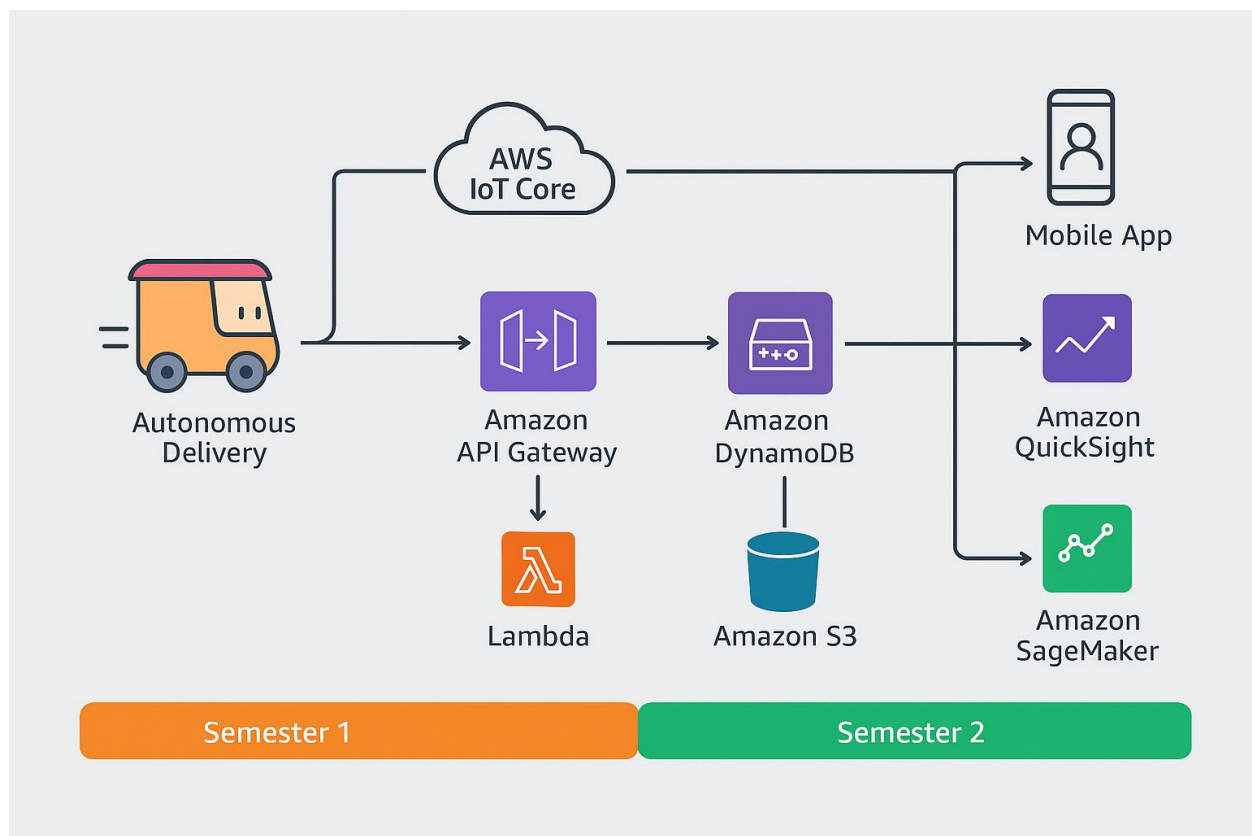
5. System Architecture (High-Level)

- **Frontend:**
 - Mobile App (customers).
 - Vendor Web App.
 - Manager Dashboard.
- **Backend (AWS):**
 - API Gateway + Lambda (serverless APIs).
 - DynamoDB + RDS (hybrid database).
 - S3 (storage) + Glacier (archival).

- IoT Core (cart-cloud communication).
- Cognito (authentication).
- CloudWatch (monitoring & logs).
- **Hardware (On-Cart):**
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6. Expected Outcomes

- A working prototype of an autonomous delivery cart system.
- Mobile app for customers, vendor portal, and manager dashboard.
- Real-time cart tracking and order workflow.
- Cloud-hosted infrastructure with scalability.
- Analytics and insights for operational decision-making.



7. Deliverables

- **Semester 1 Submission:**
 - Requirement documents, diagrams, SRS, BRD.
 - Cloud backend MVP with APIs, database, IoT setup.
 - Early mobile app prototype.
- **Semester 2 Submission:**
 - Full system integration (mobile, vendor, dashboard).
 - Data analytics & AI integration.
 - Final tested and documented solution.

8. Tools & Technologies

- **Cloud:** AWS (API Gateway, Lambda, DynamoDB, RDS, S3, IoT Core, Cognito).
- **Frontend:** Flutter (mobile), React.js (web).
- **Data Analytics:** AWS Glue, Athena, QuickSight.
- **AI/ML:** NVIDIA Jetson, TensorRT (for inference).
- **Hardware Sensors:** Azure Kinect DK, ZED, LiDAR.

9. Team Roles

- **Ahmed Sameh:** Cloud backend, software architecture, system design.
- **Youssef:** Mobile application development.

10. Timeline

- **Semester 1 (14 weeks):** MVP backend, IoT connection, early mobile app, documentation.
- **Semester 2 (14 weeks):** Dashboard, analytics, AI integration, optimization, final testing.

11. Future Vision

- Multi-cart fleet management.
- Payment integration.
- AI route optimization.
- Expansion into larger smart-city ecosystems.